WEEK 1 – Python Basics (Non-Negotiable)

Learn:

- Python syntax
- Variables & data types
- Input / output
- If–else
- Loops (for, while)
- Basic functions

Practice:

- Number guessing game
- Prime number checker
- Simple calculator

🎯 **Checkpoint**:
You can write small programs without copying.

WEEK 2 – Python Intermediate

Learn:

- Lists, tuples, sets, dictionaries
- String operations
- Functions (arguments, return)
- File handling (read/write)
- Exception handling

Practice:

- Student marks management program
- Word frequency counter
- File-based data reader

🎯 **Checkpoint**:
You can store, process, and retrieve data.

WEEK 3 – NumPy (Core for ML)

Learn:

- NumPy arrays
- Array indexing & slicing
- Shape, reshape
- Broadcasting
- Vectorized operations

Practice:

- Matrix operations
- Mean, median, std using NumPy
- Speed comparison vs Python loops

🎯 **Checkpoint**:
You understand **why NumPy is faster**.

WEEK 4 – Pandas (MOST IMPORTANT WEEK)

Learn:

- Series & DataFrame
- Reading CSV/Excel
- head(), info(), describe()
- Indexing (loc, iloc)
- Handling missing values
- Filtering & sorting

Practice:

- Clean a messy CSV
- Remove nulls, duplicates
- Generate summary reports

🎯 **Checkpoint**:
You can **clean real-world data**.

WEEK 5 – Data Visualization

Learn:

- Matplotlib basics
- Seaborn basics
- Line, bar, scatter, histogram, boxplot
- Choosing the right plot

Practice:

- Sales data visualization
- Student performance charts

🎯 **Checkpoint**:
You can **explain a dataset visually**.

WEEK 6 – Math for ML (Only What Matters)

Learn:

Linear Algebra

- Scalars, vectors, matrices
- Matrix addition & multiplication
- Dot product
- Transpose

Statistics

- Mean, median, mode
- Variance & standard deviation
- Correlation
- Normal distribution (intuition)

🎯 **Checkpoint**:
You understand **what's happening inside algorithms**.

WEEK 7 – Probability + Optimization Basics

Learn:

- Probability basics
- Conditional probability
- Bayes theorem (intuition only)
- Gradient descent (conceptual)
- Cost / loss function idea

🎯 **Checkpoint**:
You can explain **why models improve over time**.

WEEK 8 – Tools + Mini Project

Learn:

- Jupyter Notebook
- Google Colab
- Virtual environments
- Basic Git & GitHub

Mini Project:

📌 **Exploratory Data Analysis Project**

- Choose any dataset
- Clean it
- Visualize it
- Write conclusions

🎯 **Checkpoint**:
You're ready to **start Machine Learning**.